Personally implemented formulae in WannierBerri
Working handbook: geometry, covariant derivatives, and interpolation

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(Dated: Working copy, 4 October 2026)

Revision colours. Black text is retained from the original `formulae.tex`; blue text marks additions and rewritten passages. Unchanged equations remain black even when labels have been added.

This working copy consolidates the results in `formulae.tex`, `Wannier.tex`, and `QGD.tex`. It establishes common conventions for the subsequent reconciliation of the v1.7 and v1.8 calculators. Formula identities, representation assumptions, and implementation status are stated separately. The original source notes remain unchanged. Source corrections and unfinished results are recorded in Appendix A.

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I. BASIC INGREDIENTS

A. Notation, units, and scope

We use Cartesian wave-vector derivatives $\partial_a = \partial/\partial k_a$ and $\epsilon_{xyz} = +1$. Repeated Cartesian indices are summed only in explicitly contracted expressions; free indices remain free. An isolated band is denoted by n. In block formulae, $m, n \in A$ and $l, p \in B$, where the two groups are separated by a nonzero gap at the wave vector under consideration. Degeneracies within a group are allowed. Primed band indices are summed over their indicated group.

$$\Delta_{rs} = \varepsilon_r - \varepsilon_s, \qquad \int [\mathrm{d}\boldsymbol{k}] = \int_{\mathrm{BZ}} \frac{\mathrm{d}^d k}{(2\pi)^d}, \qquad e > 0, \quad q_{\mathrm{electron}} = -e. \tag{1}$$

The geometric expressions use energies and wave vectors, without silently setting $\hbar = 1$. In particular,

$$v_n^a = \frac{1}{\hbar}\partial_a \varepsilon_n, \qquad E_n^a = \partial_a \varepsilon_n. \tag{2}$$

If energy is in eV and length in Å, E_n^a is in eV Å; it is not yet a velocity in m/s. The conversion is $v_n^a = (1\,\mathrm{eV\ in\ J})(1\,\text{Å in m})E_n^a/\hbar$.

Symbol	Meaning	Relation to the source notes
Q_n^{ab}	quantum geometric tensor (QGT)	QG; called QMT in `formulae.tex`
G_n^{ab}	quantum metric	g_n^{ab} in the other two notes
$\mathcal{G}_n^{ab}$	Berry connection polarizability (BCP)	original normalization retained
D^a	$U^\dagger \partial_a U$	coefficient-space connection
$\bar{A}^a$	$U^\dagger A^{W,a} U$	rotated Wannier connection
M^a	$\bar{A}^a + \mathrm{i}D^a$	auxiliary connection in BCP formulae
$\widetilde{G}^{ab}$	energy-weighted geometric matrix	distinct from the metric G_n^{ab}
$\mathfrak{m}^a$	interband magnetic kernel	separates magnetic M from BCP M

B. Berry connection

Single band Berry connection (Abelian Berry connection):

$$A_n^i(\boldsymbol{k}) = i\langle u_n(\boldsymbol{k})|\partial_{k_i} u_n(\boldsymbol{k})\rangle \equiv i\langle u_n|\partial_i u_n\rangle \tag{3}$$

Interband Berry connection (non-Abelian Berry connection):

$$A_{nm}^i(\boldsymbol{k}) = i\langle u_n(\boldsymbol{k})|\partial_{k_i} u_m(\boldsymbol{k})\rangle \equiv i\langle u_n|\partial_i u_m\rangle \tag{4}$$

The matrix Berry connection is Hermitian, $(A^a)^\dagger = A^a$. Its full matrix transforms inhomogeneously under a wave-vector-dependent change of basis. Off-block elements transform covariantly under independent rotations within A and B.

C. Quantum geometric tensor (QGT)

Quantum geometry tensor:

$$\begin{aligned} Q_n^{ij}(\boldsymbol{k}) &= \langle \partial_{k_i} u_n(\boldsymbol{k})|(1 - |u_n(\boldsymbol{k})\rangle\langle u_n(\boldsymbol{k})|)|\partial_{k_j} u_n(\boldsymbol{k})\rangle \\ &\equiv \langle \partial_i u_n|(1 - |u_n\rangle\langle u_n|)|\partial_j u_n\rangle \\ &= \sum_{m(\neq n)} \langle \partial_i u_n|u_m\rangle\langle u_m|\partial_j u_n\rangle \\ &= \sum_{m(\neq n)} A_{nm}^i A_{mn}^j \\ &= G_n^{ij}(\boldsymbol{k}) - \frac{i}{2}\Omega_n^{ij}(\boldsymbol{k}) \end{aligned} \tag{5}$$

Quantum metric tensor:

$$G_n^{ij}(\boldsymbol{k}) = \mathrm{Re}\; Q_n^{ij}(\boldsymbol{k}) = \mathrm{Re} \sum_{m(\neq n)} A_{nm}^i A_{mn}^j \tag{6}$$

Berry curvature tensor:

$$\Omega_n^{ij}(\boldsymbol{k}) = -2\mathrm{Im}\; Q_n^{ij}(\boldsymbol{k}) = -2\mathrm{Im} \sum_{m(\neq n)} A_{nm}^i A_{mn}^j \tag{7}$$

Berry curvature vector:

$$\Omega_n^i(\boldsymbol{k}) = \frac{1}{2}\epsilon^{ijk}\Omega_n^{jk}(\boldsymbol{k}) = -\epsilon^{ijk}\mathrm{Im} \sum_{m(\neq n)} A_{nm}^j A_{mn}^k = \epsilon^{ijk}\partial_j A_n^k \tag{8}$$

The band sum in Eq. (5) requires a complete complement to n. A finite-model band sum gives the geometry of that model; missing states in a Wannier representation require the external terms of Sec. III. The metric is symmetric and positive semidefinite, whereas $\Omega_n^{ab} = -\Omega_n^{ba}$.

D. Berry connection polarizability (BCP)

Berry connection polarizability [2]:

$$\mathcal{G}_n^{ij}(\boldsymbol{k}) = \frac{1}{2} \sum_{m(\neq n)} \frac{A_{nm}^i A_{mn}^j + A_{nm}^j A_{mn}^i}{\varepsilon_n - \varepsilon_m} = \mathrm{Re} \sum_{m(\neq n)} \frac{A_{nm}^i A_{mn}^j}{\varepsilon_n - \varepsilon_m} \tag{9}$$

The normalization in Eq. (9) is retained from the original notes. Any response formula imported from another convention must first be converted to this definition; a factor of two must not be inferred from the name BCP alone. The units of Q, G, and Ω are length squared; those of $\mathcal{G}$ are length squared divided by energy.

II. COVARIANT DERIVATIVES AND WANNIER CONVENTIONS

A. Wannier and Hamiltonian gauges

With Wannier centers $\boldsymbol{t}_i$, define

$$|u_{j\boldsymbol{k}}^W\rangle = \sum_{\boldsymbol{R}} e^{\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{R}+\boldsymbol{t}_j-\boldsymbol{r})}|\boldsymbol{R}j\rangle, \tag{10}$$

$$\mathcal{H}_{ij}^W(\boldsymbol{k}) = \sum_{\boldsymbol{R}} e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_{ij}}\langle \boldsymbol{0}i|\hat{\mathcal{H}}|\boldsymbol{R}j\rangle, \qquad \boldsymbol{R}_{ij} = \boldsymbol{R} + \boldsymbol{t}_j - \boldsymbol{t}_i, \tag{11}$$

$$\mathcal{H}^W U = U\varepsilon, \qquad |u_n\rangle = \sum_j |u_j^W\rangle U_{jn}, \qquad \bar{X} = U^\dagger X^W U. \tag{12}$$

Normalization of the Bloch sums is understood. Fourier derivatives follow by multiplying every real-space term by $\mathrm{i}R_{ij}^a$, with one factor for each derivative.

For the coefficient vectors (columns of U),

$$D^a = U^\dagger \partial_a U, \qquad (D^a)^\dagger = -D^a, \qquad (D_{nm}^a)^* = -D_{mn}^a, \tag{13}$$

$$D_{ln}^a = \frac{\mathcal{H}_{ln}^{\prime a}}{\varepsilon_n - \varepsilon_l}, \quad l \in B,\ n \in A, \qquad \mathcal{H}^{\prime a} = U^\dagger(\partial_a \mathcal{H}^W)U, \tag{14}$$

$$|\partial_a u_n\rangle = \sum_j |\partial_a u_j^W\rangle U_{jn} + \sum_r |u_r\rangle D_{rn}^a, \tag{15}$$

$$A^{H,a} = \bar{A}^a + \mathrm{i}D^a \equiv M^a, \qquad A_{ij}^{W,a} = \mathrm{i}\langle u_i^W|\partial_a u_j^W\rangle. \tag{16}$$

Diagonal entries of D depend on the phase choice. Equation (14) is used only across separated blocks, never to divide by a zero gap within a degenerate group. In general, $D_{nm}^a \neq \langle u_n|\partial_a u_m\rangle$: the latter also contains the derivative of the Wannier basis.

B. Comma, ordinary, and covariant derivatives

For a specified Wannier-gauge matrix X^W, the comma derivative means

$$X^{,d} = U^\dagger(\partial_d X^W)U, \qquad \partial_d \bar{X} = X^{,d} + \bar{X}D^d - D^d\bar{X}. \tag{17}$$

Thus $X^{,d}$ is not the ordinary derivative of $\bar{X}$. In a fixed basis for an abstract operator it becomes $\langle n|\partial_d \hat{X}|m\rangle$. Such an operator-derivative matrix is covariant; it is the ordinary derivative of a transformed matrix that acquires derivatives of the gauge rotation.

Under $U \mapsto U\,\mathrm{diag}(W_A, W_B)$, $X_{AB} \mapsto W_A^\dagger X_{AB} W_B$. Using the coefficient-space connection, define [1]

$$X^{:d}_{CD} = \partial_d X_{CD} + D^d_{CC}X_{CD} - X_{CD}D^d_{DD}, \quad C, D \in \{A, B\}. \tag{18}$$

Here C, D label blocks, while the superscript d labels a Cartesian direction. The symbol X in this equation denotes a Hamiltonian-gauge matrix. For a rotated Wannier matrix the resulting block rules are

$$X^{:d}_{AB} = X^{,d}_{AB} - D^d_{AB}X_{BB} + X_{AA}D^d_{AB}, \tag{19}$$

$$X^{:d}_{BA} = X^{,d}_{BA} - D^d_{BA}X_{AA} + X_{BB}D^d_{BA}, \tag{20}$$

$$X^{:d}_{AA} = X^{,d}_{AA} - D^d_{AB}X_{BA} + X_{AB}D^d_{BA}. \tag{21}$$

For the Hamiltonian, $\mathcal{H}^{:d}_{AA} = \mathcal{H}^{,d}_{AA}$ and $\mathcal{H}^{:d}_{AB} = 0$. These statements concern rotations within a fixed Wannier representation. A connection defined using the full physical Bloch basis must be used consistently if that representation is changed.

C. Trace, product, and chain rules

The trace and product rules eliminate gauge-dependent intrablock connections:

$$\partial_d \,\mathrm{Tr}_A\, X = \mathrm{Tr}_A\, X^{:d}, \tag{22}$$

$$(X_{CD}Y_{DE})^{:d} = X^{:d}_{CD}Y_{DE} + X_{CD}Y^{:d}_{DE}. \tag{23}$$

For $\widetilde{f} = f(\mathcal{H})$, evaluation in the energy eigenbasis gives

$$\widetilde{f}^{:d}_{mn} = \mathcal{H}^{:d}_{mn} \begin{cases} f'(\varepsilon_n), & \varepsilon_m = \varepsilon_n, \\ \dfrac{f(\varepsilon_m) - f(\varepsilon_n)}{\varepsilon_m - \varepsilon_n}, & \varepsilon_m \neq \varepsilon_n. \end{cases} \tag{24}$$

The divided difference is essential when a block contains distinct energies. At zero temperature, apply the smooth-function rule before taking the appropriate distributional limit of the occupation.

For compact second derivatives, define the bilinear product expansion

$$\begin{aligned} \mathscr{P}_d(X,Y) &= X^{:d}Y + XY^{:d}, \\ \mathscr{P}_{de}(X,Y) &= X^{:de}Y + X^{:d}Y^{:e} + X^{:e}Y^{:d} + XY^{:de}, \\ X^{:de} &\equiv (X^{:d})^{:e}. \end{aligned} \tag{25}$$

Matrix order is preserved. Ordered covariant derivatives need not commute; ordinary derivatives of a smooth traced scalar do commute.

III. GAUGE-COVARIANT GEOMETRY AND WANNIER INTERPOLATION

A. Projectors and geometric matrices

Let $P = \sum_{n\in A} |u_n\rangle\langle u_n|$ and $Q = 1 - P$. For physical Bloch states, Q includes states outside a finite Wannier space. The projected geometric matrices are

$$\widetilde{F}^{ab}_{mn} = \langle \partial_a u_m | Q | \partial_b u_n \rangle, \tag{26}$$

$$\widetilde{G}^{ab} = \tfrac{1}{2}(\mathcal{H}_{AA}\widetilde{F}^{ab} + \widetilde{F}^{ab}\mathcal{H}_{AA}), \tag{27}$$

$$\widetilde{H}^{ab}_{mn} = \langle \partial_a u_m | Q\hat{\mathcal{H}}Q | \partial_b u_n \rangle. \tag{28}$$

Each transforms as $\widetilde{O} \mapsto W_A^\dagger \widetilde{O} W_A$ and obeys $\widetilde{O}^{ab}_{mn} = (\widetilde{O}^{ba}_{nm})^*$. In the Hamiltonian gauge, $\widetilde{G}^{ab}_{mn} = (\varepsilon_m + \varepsilon_n)\widetilde{F}^{ab}_{mn}/2$. The metric and curvature matrices are

$$\widetilde{g}^{ab} = \tfrac{1}{2}(\widetilde{F}^{ab} + \widetilde{F}^{ba}), \qquad \widetilde{\Omega}^c = \mathrm{i}\epsilon_{cab}\widetilde{F}^{ab}. \tag{29}$$

For $A = \{n\}$ these recover Sec. I. For a larger A, $\mathrm{Tr}_A\, \widetilde{g}$ is the metric of the whole subspace; it is generally not $\sum_{n\in A} G_n$.

For a continuum Hamiltonian with kinetic energy $(\boldsymbol{p} + \hbar\boldsymbol{k})^2/(2m_e)$ and a local, momentum-independent potential, the fourth matrix in `Wannier.tex` is

$$\widetilde{S}^{ab}_{mn} = \frac{\hbar^2}{m_e}\delta_{ab}\delta_{mn} + 2(\widetilde{G}^{ab}_{mn} - \widetilde{H}^{ab}_{mn}). \tag{30}$$

For $A = \{n\}$, $\mathrm{Re}\, \widetilde{S}^{ab}_{nn} = \partial_a\partial_b\varepsilon_n$. The free-electron term cannot be substituted for the bare Hessian of a general tight-binding or nonlocal Hamiltonian; use Sec. VI there.

B. Internal and external terms

Define Wannier-gauge matrices

$$\begin{aligned} F^{W,ab}_{ij} &= \langle \partial_a u^W_i | \partial_b u^W_j \rangle, \\ B^{W,a}_{ij} &= \mathrm{i}\langle u^W_i | \hat{\mathcal{H}} | \partial_a u^W_j \rangle, \\ H^{W,ab}_{ij} &= \langle \partial_a u^W_i | \hat{\mathcal{H}} | \partial_b u^W_j \rangle. \end{aligned} \tag{31}$$

The two-index $H^{W,ab}$ is an energy-weighted geometric matrix, not $\partial_a\partial_b\mathcal{H}^W$. Then

$$\widetilde{F}^{ab}_{\mathrm{int}} = -D^a_{AB}D^b_{BA}, \tag{32}$$

$$\widetilde{F}^{ab}_{\mathrm{ext}} = \bar{F}^{ab}_{AA} - \bar{A}^a_{AA}\bar{A}^b_{AA} + \mathrm{i}D^a_{AB}\bar{A}^b_{BA} + \mathrm{i}\bar{A}^a_{AB}D^b_{BA}, \tag{33}$$

$$\widetilde{F}^{ab} = \widetilde{F}^{ab}_{\mathrm{int}} + \widetilde{F}^{ab}_{\mathrm{ext}}. \tag{34}$$

Here B is the complement of A within the represented Wannier space; $\bar{F}$ retains the additional basis-derivative information. Keeping only $\widetilde{F}_{\mathrm{int}}$ corresponds to the fixed-basis Hamiltonian geometry.

For $\widetilde{H}$, assuming the represented active states are eigenstates of the physical Hamiltonian (or lie in the exact frozen subspace),

$$\begin{aligned} \widetilde{H}^{ab}_{\mathrm{int}} &= -D^a_{AB}\mathcal{H}_{BB}D^b_{BA}, \\ \widetilde{H}^{ab}_{\mathrm{ext}} &= \bar{H}^{ab}_{AA} - \bar{A}^a_{AA}\mathcal{H}_{AA}\bar{A}^b_{AA} + \mathrm{i}D^a_{AB}\bar{B}^b_{BA} + \mathrm{i}(\bar{B}^a)^\dagger_{AB}D^b_{BA}. \end{aligned} \tag{35}$$

Outside that exact subspace, projection and disentanglement errors must be assessed before interpreting the interpolation as a physical matrix element.

A useful sign check is

$$\Omega^{ab}_{A,\mathrm{int}} = 2\,\mathrm{Im}\,\mathrm{Tr}_A(D^a_{AB}D^b_{BA}), \qquad \widetilde{\Omega}^z = \mathrm{i}(\widetilde{F}^{xy} - \widetilde{F}^{yx}). \tag{36}$$

C. First and second geometric derivatives

The internal first derivative is

$$\widetilde{F}^{ab:d}_{\mathrm{int}} = -D^{a:d}_{AB}D^b_{BA} - D^a_{AB}D^{b:d}_{BA}. \tag{37}$$

The external first derivative contains all seven product-rule terms:

$$\begin{aligned} \widetilde{F}^{ab:d}_{\mathrm{ext}} = {} & \bar{F}^{ab:d}_{AA} - \bar{A}^{a:d}_{AA}\bar{A}^b_{AA} - \bar{A}^a_{AA}\bar{A}^{b:d}_{AA} \\ & + \mathrm{i}D^{a:d}_{AB}\bar{A}^b_{BA} + \mathrm{i}D^a_{AB}\bar{A}^{b:d}_{BA} + \mathrm{i}\bar{A}^{a:d}_{AB}D^b_{BA} + \mathrm{i}\bar{A}^a_{AB}D^{b:d}_{BA}. \end{aligned} \tag{38}$$

All second-order terms are retained compactly by Eq. (25):

$$\begin{aligned}\widetilde{F}^{ab:de}_{\rm int} &= -\mathscr{P}_{de}(D^a_{AB}, D^b_{BA}),\\ \widetilde{F}^{ab:de}_{\rm ext} &= \bar{F}^{ab:de}_{AA} - \mathscr{P}_{de}(\bar{A}^a_{AA}, \bar{A}^b_{AA})\\ &\quad + \mathrm{i}\mathscr{P}_{de}(D^a_{AB}, \bar{A}^b_{BA}) + \mathrm{i}\mathscr{P}_{de}(\bar{A}^a_{AB}, D^b_{BA}).\end{aligned} \tag{39}$$

Consequently, for $s = d$ or de,

$$\widetilde{g}^{ab:s} = \tfrac{1}{2}(\widetilde{F}^{ab:s} + \widetilde{F}^{ba:s}), \qquad \widetilde{\Omega}^{c:s} = \mathrm{i}\epsilon_{cab}\widetilde{F}^{ab:s}. \tag{40}$$

Taking Tr_A gives the ordinary first or second derivative of the corresponding subspace trace. The same product rule differentiates Eqs. (27) and (35); derivatives of every Hamiltonian factor are required.

IV. BCP AND ENERGY-DENOMINATOR DERIVATIVES

A. A symmetric block construction

For off-block elements define M by Eq. (16) and

$$N^a_{rs} = \frac{M^a_{rs}}{\Delta_{rs}}, \qquad (M^a_{rs})^* = M^a_{sr}, \qquad (N^a_{rs})^* = -N^a_{sr}. \tag{41}$$

The finite-model energy-weighted tensor and BCP are

$$\widetilde{\mathcal{F}}^{ab} = \tfrac{1}{2}(N^a_{AB}M^b_{BA} - M^a_{AB}N^b_{BA}), \tag{42}$$

$$\widetilde{\mathcal{G}}^{ab} = \tfrac{1}{2}(\widetilde{\mathcal{F}}^{ab} + \widetilde{\mathcal{F}}^{ba}). \tag{43}$$

In the energy eigenbasis,

$$\widetilde{\mathcal{F}}^{ab}_{mn} = \frac{1}{2}\sum_{l\in B} M^a_{ml}M^b_{ln}\left(\frac{1}{\Delta_{ml}} + \frac{1}{\Delta_{nl}}\right). \tag{44}$$

For a single active band and a complete model complement, $\widetilde{\mathcal{G}}^{ab}_{nn} = \mathcal{G}^{ab}_n$. A calculation using only M within a truncated Wannier space does not restore energy-denominator contributions from all excluded physical bands.

Let $T^{ab} = N^a_{AB}M^b_{BA} - M^a_{AB}N^b_{BA}$. Hermiticity gives $(T^{ab})^\dagger = T^{ba}$, so the code-friendly form is

$$\widetilde{\mathcal{G}}^{ab} = \tfrac{1}{4}[T^{ab} + (T^{ab})^\dagger]. \tag{45}$$

This explains the division by four after band-index Hermitian symmetrization in the archived `BCP` class.

B. Connection derivatives and the general block rule

The first derivative of the off-block coefficient connection is

$$\begin{aligned}D^{a:d}_{ln} = \frac{1}{\varepsilon_n - \varepsilon_l}\big[&\mathcal{H}^{;ad}_{ln} + \mathcal{H}^{;a}_{lp}D^d_{pn} - D^d_{lm}\mathcal{H}^{;a}_{mn}\\ &+ \mathcal{H}^{;d}_{lp}D^a_{pn} - D^a_{lm}\mathcal{H}^{;d}_{mn}\big],\end{aligned} \tag{46}$$

where $m \in A$ and $p \in B$ are summed. Also $M^{a:d} = \bar{A}^{a:d} + \mathrm{i}D^{a:d}$, with the corresponding relation at second order.

To differentiate an energy denominator covariantly, define the Sylvester operator

$$\mathscr{L}_{CD}(Y) = \mathcal{H}_{CC}Y - Y\mathcal{H}_{DD}. \tag{47}$$

Its inverse exists for separated spectra. Equation (41) is the Hamiltonian-gauge evaluation of $\mathscr{L}_{CD}(N^a) = M^a$. For $\mathscr{L}_{CD}(Y) = R$, product differentiation gives

$$\mathscr{L}_{CD}(Y^{:d}) = R^{:d} - \mathcal{H}^{:d}_{CC}Y + Y\mathcal{H}^{:d}_{DD}, \tag{48}$$

$$\begin{aligned}\mathscr{L}_{CD}(Y^{:de}) = R^{:de} &- \mathcal{H}^{:de}_{CC}Y - \mathcal{H}^{:d}_{CC}Y^{:e} - \mathcal{H}^{:e}_{CC}Y^{:d}\\ &+ Y^{:d}\mathcal{H}^{:e}_{DD} + Y^{:e}\mathcal{H}^{:d}_{DD} + Y\mathcal{H}^{:de}_{DD}.\end{aligned} \tag{49}$$

Set $(Y, R) = (N^a, M^a)$ for BCP denominators, or $(Y, R) = (D^a_{BA}, -\mathcal{H}^a_{BA})$ for coefficient connections. These identities preserve intrablock matrix products and supply a general reference for auditing the multiband implementation.

C. Isolated-band quotient formulae

For individual nondegenerate eigenlines, write the phase-covariant derivative as $;d$, to distinguish it from a multiband block derivative. Let $\Delta_{rs,d} = E^d_r - E^d_s$ and $\Delta_{rs,de} = \partial_e\partial_d(\varepsilon_r - \varepsilon_s)$. Then

$$N^{a;d}_{rs} = \frac{M^{a;d}_{rs}}{\Delta_{rs}} - \frac{M^a_{rs}\Delta_{rs,d}}{\Delta^2_{rs}}, \tag{50}$$

$$N^{a;de}_{rs} = \frac{M^{a;de}_{rs}}{\Delta_{rs}} - \frac{M^{a;d}_{rs}\Delta_{rs,e} + M^{a;e}_{rs}\Delta_{rs,d} + M^a_{rs}\Delta_{rs,de}}{\Delta^2_{rs}} + \frac{2M^a_{rs}\Delta_{rs,d}\Delta_{rs,e}}{\Delta^3_{rs}}. \tag{51}$$

These are the compact quotient formulae in `Wannier.tex`. They must not replace Eqs. (48)–(49) for arbitrary multiband rotations. The archived `DerBCP` uses diagonal velocity differences; its general-block equivalence remains a reconciliation item.

D. BCP dipole and quadrupole kernels

With either consistently evaluated block derivatives or isolated-band derivatives,

$$\widetilde{\mathcal{F}}^{ab:d} = \tfrac{1}{2}\big[N^{a:d}_{AB}M^b_{BA} + N^a_{AB}M^{b:d}_{BA} - M^{a:d}_{AB}N^b_{BA} - M^a_{AB}N^{b:d}_{BA}\big], \tag{52}$$

$$\widetilde{\mathcal{F}}^{ab:de} = \tfrac{1}{2}\big[\mathscr{P}_{de}(N^a_{AB}, M^b_{BA}) - \mathscr{P}_{de}(M^a_{AB}, N^b_{BA})\big], \tag{53}$$

$$\widetilde{\mathcal{G}}^{ab:s} = \tfrac{1}{2}(\widetilde{\mathcal{F}}^{ab:s} + \widetilde{\mathcal{F}}^{ba:s}), \qquad s = d, de. \tag{54}$$

The two factors of $1/2$ in this construction are independent. For a smooth subspace, $\mathrm{Tr}_A\,\widetilde{\mathcal{G}}^{ab:d}$ and $\mathrm{Tr}_A\,\widetilde{\mathcal{G}}^{ab:de}$ are the dipole and quadrupole geometric kernels before occupations and response prefactors are applied.

V. EXPLICIT BAND-SUM AND MATRIX FORMULAE FROM QGD

A. Domain and matrix ingredients

This section applies to a smooth Hamiltonian in a fixed orthonormal basis, with nondegenerate bands and nonzero denominators. It gives an independent representation of the internal geometry. Define energy-derivative matrices, rather than physical velocities,

$$V^a_{nm} = \langle n|\partial_a\mathcal{H}|m\rangle, \qquad \Gamma^{ab}_{nm} = \langle n|\partial_a\partial_b\mathcal{H}|m\rangle,$$

$$(K^a_q)_{nm} = \begin{cases} V^a_{nm}/\Delta^q_{nm}, & n \neq m, \\ 0, & n = m, \end{cases} \qquad (V^a_F)_{nm} = \delta_{nm}E^a_n. \tag{55}$$

Division by Δ^q is elementwise and excludes the diagonal. Products of K_q are ordinary matrix products. In this convention $D^a_{nm} = -(K^a_1)_{nm}$ and $A^a_{nm} = -\mathrm{i}(K^a_1)_{nm}$ for $n \neq m$. Thus K_1 alone is not the interband Berry connection.

The QGT can be evaluated by either expression

$$Q^{ab}_n = \sum_{m\neq n} \frac{V^a_{nm}V^b_{mn}}{\Delta^2_{nm}} = -[K^a_1K^b_1]_{nn}. \tag{56}$$

Only the diagonal of the matrix product is asserted to equal the single-band QGT. Its off-diagonal entries are not the general projected matrix of Eq. (26).

B. QGD-I and QGD-II

Write $\partial_c Q^{ab}_n = \mathcal{I}^{ab;c}_n + \mathcal{J}^{ab;c}_n$. The one-sum contribution is

$$\mathcal{I}^{ab;c}_n = \sum_{m\neq n}\left[\frac{\Gamma^{ca}_{nm}V^b_{mn} + V^a_{nm}\Gamma^{cb}_{mn}}{\Delta^2_{nm}} - \frac{2V^a_{nm}V^b_{mn}(E^c_n - E^c_m)}{\Delta^3_{nm}}\right]. \tag{57}$$

The two-sum contribution is

$$\mathcal{J}_n^{ab;c} = \sum_{m\neq n}\Bigg[\sum_{p\neq n}\frac{V_{np}^c V_{pm}^a V_{mn}^b + V_{nm}^a V_{mp}^b V_{pn}^c}{\Delta_{nm}^2\Delta_{np}} + \sum_{p\neq m}\frac{V_{np}^a V_{pm}^c V_{mn}^b + V_{nm}^a V_{mp}^c V_{pn}^b}{\Delta_{nm}^2\Delta_{mp}}\Bigg]. \tag{58}$$

The corresponding compact matrix expressions are

$$\mathcal{I}_n^{ab;c} = \left[\Gamma^{ca}K_2^b + K_2^a\Gamma^{cb} - 2[V_F^c, K_1^a]K_2^b\right]_{nn}, \tag{59}$$

$$\mathcal{J}_n^{ab;c} = \left[[K_1^c, V^a]K_2^b + K_2^a[K_1^c, V^b]\right]_{nn}. \tag{60}$$

No real part is taken until selecting the desired observable:

$$\partial_c G_n^{ab} = \mathrm{Re}\,\partial_c Q_n^{ab}, \qquad \partial_c\Omega_n^{ab} = -2\,\mathrm{Im}\,\partial_c Q_n^{ab}. \tag{61}$$

Equations (57)–(60) retain the one-sum/two-sum distinction in `QGD.tex` without repeating the intermediate bra-ket expansions.

C. A complete quotient identity for the second derivative

The QGQ expansions in the source notes are incomplete. For a compact check of any completed expansion, define the phase-invariant pair $Z_{nm}^{ab} = V_{nm}^a V_{mn}^b$ for $m \neq n$. Differentiating $Q_n^{ab} = \sum_{m\neq n} Z_{nm}^{ab}/\Delta_{nm}^2$ twice gives

$$\partial_d\partial_c Q_n^{ab} = \sum_{m\neq n}\Bigg[\frac{\partial_d\partial_c Z_{nm}^{ab}}{\Delta_{nm}^2} - \frac{2\big[(\partial_c Z_{nm}^{ab})\Delta_{nm,d} + (\partial_d Z_{nm}^{ab})\Delta_{nm,c} + Z_{nm}^{ab}\Delta_{nm,cd}\big]}{\Delta_{nm}^3} + \frac{6Z_{nm}^{ab}\Delta_{nm,c}\Delta_{nm,d}}{\Delta_{nm}^4}\Bigg]. \tag{62}$$

In particular, the last coefficient is $+6$ in this scalar band-sum convention. The numerator derivatives include changes of both eigenstates and operators. For an arbitrary operator matrix X, they can be generated using

$$X_{nm}^{;c} = \langle n|\partial_c\hat{X}|m\rangle + [K_1^c, X]_{nm}, \qquad (K_q^a)_{nm}^{;c} = \frac{(V^a)_{nm}^{;c}}{\Delta_{nm}^q} - q\frac{V_{nm}^a\Delta_{nm,c}}{\Delta_{nm}^{q+1}}, \quad n\neq m. \tag{63}$$

Here $;c$ removes only eigenline phase connections, with the same choice as in Sec. IV. Differentiating Γ includes $\langle n|\partial_c\partial_a\partial_b\mathcal{H}|m\rangle$. It cannot be omitted for a general tight-binding Hamiltonian. This identity is a complete differentiation rule, not a claim that the partial QGQ matrix expansions in the source notes are complete.

VI. ENERGY DERIVATIVES AND DRUDE BUILDING BLOCKS

Define the rotated bare derivatives

$$V^a = \mathcal{H}^{,a}, \quad W^{ab} = \mathcal{H}^{,ab}, \quad Y^{abc} = \mathcal{H}^{,abc}, \quad Z^{abcd} = \mathcal{H}^{,abcd}. \tag{64}$$

Here Z^{abcd} is a four-derivative Hamiltonian matrix, distinct from the band-pair numerator Z_{nm}^{ab} in Sec. V. For example,

$$V_{AA}^{a:b} = W_{AA}^{ab} + V_{AB}^a D_{BA}^b - D_{AB}^b V_{BA}^a, \tag{65}$$

$$W_{AA}^{ab:c} = Y_{AA}^{abc} + W_{AB}^{ab}D_{BA}^c - D_{AB}^c W_{BA}^{ab}, \tag{66}$$

$$V_{AA}^{a:bc} = W_{AA}^{ab:c} + V_{AB}^{a:c}D_{BA}^b + V_{AB}^a D_{BA}^{b:c} - D_{AB}^{b:c}V_{BA}^a - D_{AB}^b V_{BA}^{a:c}. \tag{67}$$

A further product derivative yields

$$V_{AA}^{a:bcd} = W_{AA}^{ab:cd} + V_{AB}^{a:cd}D_{BA}^b + V_{AB}^{a:c}D_{BA}^{b:d} + V_{AB}^{a:d}D_{BA}^{b:c} + V_{AB}^a D_{BA}^{b:cd} - D_{AB}^{b:cd}V_{BA}^a - D_{AB}^{b:c}V_{BA}^{a:d} - D_{AB}^{b:d}V_{BA}^{a:c} - D_{AB}^b V_{BA}^{a:cd}. \tag{68}$$

For a smooth isolated group,

$$\partial_a\partial_b \sum_{n\in A}\varepsilon_n = \mathrm{Tr}_A\, V^{a:b}, \quad \partial_a\partial_b\partial_c \sum_{n\in A}\varepsilon_n = \mathrm{Tr}_A\, V^{a:bc}, \quad \partial_a\partial_b\partial_c\partial_d \sum_{n\in A}\varepsilon_n = \mathrm{Tr}_A\, V^{a:bcd}. \tag{69}$$

For a single isolated band the inverse effective mass is $(m_n^{-1})^{ab} = \hbar^{-2}\partial_a\partial_b\varepsilon_n$. The operator Hessian W_{nn}^{ab} differs from the band-energy Hessian:

$$\partial_a\partial_b\varepsilon_n = W_{nn}^{ab} + 2\,\mathrm{Re}\sum_{m\neq n}\frac{V_{nm}^a V_{mn}^b}{\Delta_{nm}}. \tag{70}$$

Thus the orbital-quadrupole derivative of an operator Hessian must not be replaced by `Der3E`. They correspond to different objects.

VII. ORBITAL MAGNETIC QUANTITIES

A. Orbital moment and zero-temperature magnetization

With the charge convention in Eq. (1),

$$\widetilde{m}^c = \frac{\mathrm{i}e}{2\hbar}\epsilon_{cab}(\widetilde{G}^{ab} - \widetilde{H}^{ab}), \tag{71}$$

$$\widetilde{\mathcal{M}}^c(\mu) = -\frac{\mathrm{i}e}{2\hbar}\epsilon_{cab}(\widetilde{H}^{ab} + \widetilde{G}^{ab} - 2\mu\widetilde{F}^{ab}). \tag{72}$$

The first is the symmetrized orbital-moment matrix; for an exactly degenerate active group it reduces to the usual moment matrix. For a single band, $\mathcal{M}_n^c = m_n^c + (e/\hbar)(\mu-\varepsilon_n)\Omega_n^c$. At zero temperature, using the occupied projector at each wave vector,

$$M_{\mathrm{orb}}^c = \int [\mathrm{d}\boldsymbol{k}]\,\mathrm{Tr}_{\mathrm{occ}}\,\widetilde{\mathcal{M}}^c(\mu). \tag{73}$$

At fixed μ, the derivative of Eq. (72) is

$$(\widetilde{\mathcal{M}}^c)^{:d} = -\frac{\mathrm{i}e}{2\hbar}\epsilon_{cab}(\widetilde{H}^{ab:d} + \widetilde{G}^{ab:d} - 2\mu\widetilde{F}^{ab:d}). \tag{74}$$

Every projector derivative in $\widetilde{H}$ and $\widetilde{G}$ is included through the interpolation and product rules. Keeping only $\langle\partial_a u_m|Q(\partial_d\hat{\mathcal{H}})Q|\partial_b u_n\rangle$ is insufficient.

B. Orbital magnetic quadrupole: four-term reference

To avoid the duplicated prefactors in `Wannier.tex`, this subsection uses the $e=\hbar=1$ convention of Gao and Xiao [3]. It supplies the reduced kernel for comparison with the archive, not a new SI conversion. For an isolated band define

$$\mathfrak{m}_{mn}^j = \frac{1}{2}\epsilon_{jab}\sum_{p\neq n}(v_{mp}^a + v_n^a\delta_{mp})A_{pn}^b, \tag{75}$$

$$\Gamma_n^{ib} = \langle u_n|\partial_i\partial_b\hat{\mathcal{H}}|u_n\rangle, \qquad g_n^{ib} \equiv G_n^{ib}. \tag{76}$$

Here $v^a = \partial_a\hat{\mathcal{H}}$ in this subsection's units. Define four band kernels,

$$C_n^{ij} = \frac{2}{3}\,\mathrm{Re}\sum_{m\neq n}A_{nm}^i\mathfrak{m}_{mn}^j, \qquad I_n^{ij} = -\frac{1}{12}\epsilon_{abj}\partial_a\Gamma_n^{ib},$$

$$L_n^{ij} = -2\,\mathrm{Re}\sum_{m\neq n}\frac{A_{nm}^i\mathfrak{m}_{mn}^j}{\Delta_{nm}}, \qquad K_n^{ij} = -\frac{1}{3}\epsilon_{abj}\partial_a G_n^{ib}. \tag{77}$$

The quadrupole density is

$$\mathcal{Q}^{ij} = \int [\mathrm{d}\boldsymbol{k}] \sum_n \left\{ f_n (C_n^{ij} + I_n^{ij}) + \Phi_n (L_n^{ij} + K_n^{ij}) \right\}, \tag{78}$$

where

$$f_n = \frac{1}{1 + e^{(\varepsilon_n - \mu)/(k_B T)}}, \qquad \Phi_n = -k_B T \ln\left(1 + e^{(\mu - \varepsilon_n)/(k_B T)}\right). \tag{79}$$

At $T = 0$, $f_n = \Theta(\mu - \varepsilon_n)$ and $\Phi_n = (\varepsilon_n - \mu) f_n$. The source-note labels ME1, W, ME2, QM correspond respectively to the archived v1.8 labels CL, IMD, ME, QMD.

The physical operator Hessian in Eq. (77) equals the fixed-basis matrix W of Sec. VI in that representation. With a moving Wannier basis, operator derivatives and derivatives of $\mathcal{H}^W$ require additional basis-connection terms. This distinction must be checked when comparing the archived `DerWln` implementation to the physical quadrupole formula.

The archived v1.8 calculator uses $(E - \mu) f$ for ME and QMD and defaults to `constant_factor=1`. Its weighting agrees with Eq. (79) at zero temperature, and does not by itself implement the finite-temperature grand potential. No additional $e/(2\hbar)$ factor should be inserted into Eq. (78) while retaining the convention in Eq. (75). Conversion back to physical units must be derived for the full expression.

VIII. BRIDGE TO THE V1.7 AND V1.8 CALCULATOR CATALOGUE

A. Fermi-sea and Fermi-surface kernels

For a smooth periodic band quantity X_n and vanishing boundary terms,

$$\int [\mathrm{d}\boldsymbol{k}] \sum_n f_n \partial_a X_n = -\int [\mathrm{d}\boldsymbol{k}] \sum_n f_n' E_n^a X_n, \tag{80}$$

$$\int [\mathrm{d}\boldsymbol{k}] \sum_n f_n \partial_a \partial_d X_n = -\int [\mathrm{d}\boldsymbol{k}] \sum_n f_n' E_n^a \partial_d X_n, \tag{81}$$

where $f_n' = \partial f / \partial \varepsilon|_{\varepsilon_n}$ is negative. A surface kernel defined with $-f'$ has the same sign as the sea integral. The code's occupation-derivative convention and outer factor must therefore be checked together. At band crossings, use smooth group traces and the matrix occupation rule rather than applying isolated-band formulae blindly.

B. Tensor axes and formula-class index

The following table records geometric kernel axes, excluding any leading wave-vector and chemical-potential axes. It is a navigation aid for the subsequent full version reconciliation.

Formula class	Geometric kernel	Raw Cartesian axes
`Metric`	G^{ab} or subspace metric	(a, b)
`DerMetric`	$G^{ab:d}$	(a, b, d)
`BCP`	$\mathcal{G}^{ab}$	(a, b)
`DerBCP`	$\mathcal{G}^{ab:d}$	(a, b, d)
`Der2BCP`	$\mathcal{G}^{ab:de}$	(a, b, d, e)
`DerOmega`, `Der2Omega`	$\Omega^{c:d}$, $\Omega^{c:de}$	(c, d), (c, d, e)
`VelBCP`	$E^c \mathcal{G}^{ab}$	(c, a, b)
`VelDerBCP`	$E^c \mathcal{G}^{ab:d}$	(c, a, b, d)
`VelDerOmega`	$E^a \Omega^{c:d}$	(a, c, d)
`InvMass`, `Der3E`, `Der4E`	band-energy derivatives	derivative order
`Qorb_GaoXiao_*`	Eq. (78)	(i, j)

For a common dipole convention (d, a, b), transpose the three axes of `DerBCP` from (a, b, d) to (d, a, b). For a common quadrupole convention (d, a, b, e), the corresponding permutation of `Der2BCP` is $(2, 0, 1, 3)$ in zero-based tensor-axis notation. Only the trailing tensor axes should be permuted when leading axes are present.

C. Items for the subsequent complete version reconciliation

The archived source and `CHANGE_REPORT.md` identify the following scope:

- v1.7 adds metric, BCP dipole/quadrupole, Berry-curvature quadrupole, fourth-energy-derivative, and early orbital-quadrupole calculators.
- v1.8 retains several formula classes but does not expose the complete v1.7 calculator set. It adds the four-term Gao–Xiao quadrupole and `Beta_GaoXiao` thermoelectric response.
- `Der2BCP_test` in v1.7 raises `NotImplementedError`. The production `Der2BCP` also carries a bug-related TODO in the source; a class name or a completed symbolic identity is not evidence of numerical validation.
- BCP denominator derivatives, active/complement choices, tensor axes, occupation signs, and SI prefactors need class-by-class comparison. Group-metric traces and sums of isolated-band metrics must be distinguished.

The thermoelectric response and the full nonlinear conductivity prefactors are outside the completed source-note consolidation here. They should be added as separate catalogue entries with their field convention, thermal kernel, component symmetrization, and version-specific code mapping.

Appendix A: Corrections and retained evidence boundaries

1. **Connection and derivative notation.** The source notes used the same state notation for Bloch functions and columns of U. Equations (13)–(16) distinguish them. The ordinary derivative $\partial_d X$ was also confused with the operator/comma derivative $X_{,d}$; Sec. II B repairs that distinction. The right-block gauge transformation has left index n, not the unrelated index m.
2. **Curvature and Hermiticity.** The anti-Hermitian relation is $(D^a_{nm})^* = -D^a_{mn}$. The z-curvature contraction uses $xy-yx$, not $xy-yz$. The internal trace curvature has the positive sign in Eq. (36). The $+2\,\mathrm{Im}\,Q$ convention in `QGD.tex` is replaced consistently by Eq. (6).
3. **BCP factors and denominator derivatives.** The metric symmetrization and the symmetric energy-denominator construction each contribute a factor of $1/2$. An expanded derivative in `Wannier.tex` lost one of these factors and repeated terms instead of exchanging all a, b indices. Equations (52)–(54) retain the two operations explicitly. The Sylvester identities supplement the single-band quotient rule without assuming diagonal intrablock velocity matrices.
4. **Incomplete QGQ expansion.** `Wannier.tex` leaves its part III empty, and `QGD.tex` repeats the part-I label on a different group of terms. These partial expressions are not reproduced as a complete result. Equation (62) records the full quotient identity, including the $+6$ denominator term and the band-energy Hessian difference.
5. **Orbital terms.** The energy-weighted Wannier matrix uses derivatives a, b, not b, b. Orbital-moment, magnetization, and interband magnetic-kernel symbols are separated. The original quadrupole expression mixed an outer $e/(2\hbar)$ with a magnetic matrix already containing that factor. Section VII B uses a single explicit reduced-unit convention.
6. **Exploratory derivations.** The unfinished ME1 projector manipulations, partial magnetization derivatives, and terminal Green-function identities are retained in the original notes. They are not needed for the reusable formula chain in this handbook. In particular, differentiation of a projected product must include all projector derivatives; differentiating only its explicit Hamiltonian is incomplete.

Appendix B: Consistency checks for later calculator validation

Useful algebraic checks include

$$\begin{aligned} Q^{ab}_n &= (Q^{ba}_n)^*, \qquad G^{aa}_n \geq 0, \\ \widetilde{F}^{ab:s}_{mn} &= (\widetilde{F}^{ba:s}_{nm})^*, \qquad \mathcal{G}^{ab}_n = \mathcal{G}^{ba}_n, \\ \partial_d\partial_c X_n &= \partial_c\partial_d X_n \quad \text{for a smooth gauge-invariant scalar } X_n. \end{aligned} \tag{B1}$$

Compare the QGD band sums, matrix products, and centered finite differences on a gapped multiband model. Test the BCP block denominator identities under independent unitary rotations in both blocks. For Brillouin-zone integrals, compare sea/surface kernels only after matching axes, signs, units, occupations, and integration normalization. These checks do not replace convergence tests on the eventual material model.

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[2] Y. Gao, S. A. Yang, and Q. Niu, Field Induced Positional Shift of Bloch Electrons and Its Dynamical Implications, Phys. Rev. Lett. **112**, 166601 (2014).

[3] Y. Gao and D. Xiao, Orbital magnetic quadrupole moment and nonlinear anomalous thermoelectric transport, Phys. Rev. B **98**, 060402(R) (2018).